

Approach to EEG Reading

Let's start. I recommend this stepwise approach. The first step is the same as with any test, confirm the *patient's identity*. Be sure the EEG you are about to read belongs to the correct patient. Additionally, note three important points before reading. The technologist will usually mark these out for the reader.

1. *Age*: The EEG is relatively stable during adulthood. However, certain normal variants are common in young adults. A slight slowing of the posterior-dominant rhythm may be considered within normal after the age of 80, but it must remain within alpha range.
2. *State(s)*: The EEG varies with state changes; it is important to note if the patient was awake, drowsy, or asleep during the record. Furthermore, epileptic discharges are accentuated during drowsiness and sleep. If the patient is confused or comatose, then the EEG is reflective of that state. More on this later.
3. *Skull defect*: The skull serves as a natural filter of higher frequencies. EEG activity over a skull defect such as a craniotomy is typically sharper, faster, and of higher amplitude compared to the rest of the skull. This is called a breach effect. *To avoid overcall, the sharpness of a waveform must be interpreted with caution over a skull defect* [1]. Most epileptic discharges are negative (upward) but with the alteration of neuronal dipoles from brain surgery, such as a temporal lobectomy, the discharges may be positive (downward) [2].

Next, open to view the record.

Before interpreting the EEG activity, *check the settings*. These are usually placed on the top in most digital displays. They include the *recording parameters* (*filters, sensitivity, paper speed, or time base*), the *montage*, and the *calibration signal*. If you have read the first part of the book, you are already well versed with each of these terms.

Now, let's get to the exciting part: the brain waves!